# Lab 3 - Data tidying and joining

LAB

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# Introduction

In this lab you'll build the data wrangling and visualization skills you've developed so far and data tidying and joining to your repertoire.

> **Note**
>
> This lab assumes you've completed the labs so far and doesn't repeat setup and overview content from those labs. If you have not yet done those, you should go back and review the previous labs before starting on this one.

## Learning objectives

By the end of the lab, you will...

- Be able to pivot/reshape data using `tidyr`
- Continue developing your data wrangling skills using `dplyr`
- Build on your mastery of data visualizations using `ggplot2`
- Get more experience with data science workflow using R, RStudio, Git, and GitHub
- Further your reproducible authoring skills with Quarto
- Improve your familiarity with version control using Git and GitHub

## Getting started

Log in to RStudio, clone your `lab-3` repo from GitHub, open your `lab-3.qmd` document, and get started!

> **Click here if you prefer to see step-by-step instructions**

> **Important**
>
> If you run into any issues with the first steps outlined above, flag a TA for help before proceeding.

## Packages

In this lab we will work with the **tidyverse** package, which is a collection of packages for doing data analysis in a "tidy" way.

```
library(tidyverse)
```

**Render** the document which loads this package with the `library()` function.

## Guidelines

As we've discussed in lecture, your plots should include an informative title, axes should be labeled, and careful consideration should be given to aesthetic choices.

In addition, the code should all the code should be be able to be read (not run off the page) when you render to PDF. Make sure that is the case, and add line breaks where the code is running off the page.[1]

> **Note**
>
> Continuing to develop a sound workflow for reproducible data analysis is important as you complete the lab and other assignments in this course. There will be periodic reminders in this assignment to remind you to **render, commit, and push** your changes to GitHub. You should have at least 3 commits with meaningful commit messages by the end of the assignment.

# Questions

## Part 1

### Inflation across the world

For this part of the analysis you will work with inflation data from various countries in the world over the last 30 years.

```
country_inflation <- read_csv("data/country-inflation.csv")
```

## Question 1

Get to know the data.

a. `glimpse()` at the `country_inflation` data frame and answer the following questions based on the output. How many rows does `country_inflation` have and what does each row represent? How many columns does `country_inflation` have and what does each column represent?

```
glimpse(country_inflation)
```

```
Rows: 44
Columns: 30
```

```
$ country <chr> "Australia", "Austria", "Belgium", "Canada", "Czech Republic",…
$ `1993`  <dbl> 1.753653, 3.631786, 2.754426, 1.865079, 20.813026, 1.257862, 2…
$ `1994`  <dbl> 1.9696348, 2.9534065, 2.3775445, 0.1655629, 10.0394242, 1.9920…
$ `1995`  <dbl> 4.6277666, 2.2433638, 1.4679612, 2.1487603, 8.9905306, 2.08360…
$ `1996`  <dbl> 2.6153846, 1.8609741, 2.0770246, 1.5705311, 8.7587747, 2.12629…
$ `1997`  <dbl> 0.2248876, 1.3059833, 1.6281605, 1.6212164, 8.5961567, 2.18216…
$ `1998`  <dbl> 0.8601346, 0.9224669, 0.9492503, 0.9959425, 10.6983655, 1.8456…
$ `1999`  <dbl> 1.4831294, 0.5689865, 1.1208482, 1.7348430, 2.1354484, 2.49779…
$ `2000`  <dbl> 4.4574351, 2.3448684, 2.5445178, 2.7194400, 3.7753883, 2.90327…
$ `2001`  <dbl> 4.407135, 2.650000, 2.469258, 2.525120, 4.662676, 2.337875, 2.…
$ `2002`  <dbl> 2.981575, 1.810359, 1.645214, 2.258394, 1.902981, 2.424437, 1.…
$ `2003`  <dbl> 2.7325960, 1.3555538, 1.5889640, 2.7585632, 0.1187392, 2.07507…
$ `2004`  <dbl> 2.3432552, 2.0612068, 2.0972831, 1.8572587, 2.7601078, 1.15435…
$ `2005`  <dbl> 2.6918317, 2.2991377, 2.7814326, 2.2135520, 1.8570979, 1.81781…
$ `2006`  <dbl> 3.555288, 1.441547, 1.791208, 2.002025, 2.533993, 1.924221, 1.…
$ `2007`  <dbl> 2.3276113, 2.1685559, 1.8230563, 2.1383840, 2.8531244, 1.69326…
$ `2008`  <dbl> 4.350299, 3.215951, 4.489444, 2.370271, 6.358664, 3.416268, 4.…
$ `2009`  <dbl> 1.771117e+00, 5.063094e-01, -5.314567e-02, 2.994668e-01, 1.019…
$ `2010`  <dbl> 2.9183400, 1.8135317, 2.1892992, 1.7768715, 1.4727273, 2.31092…
$ `2011`  <dbl> 3.303850, 3.286583, 3.532082, 2.912135, 1.917219, 2.758682, 3.…
$ `2012`  <dbl> 1.7627802, 2.4856751, 2.8396634, 1.5156782, 3.2876231, 2.39791…
$ `2013`  <dbl> 2.44988864, 2.00015749, 1.11309594, 0.93829190, 1.43829787, 0.…
$ `2014`  <dbl> 2.48792271, 1.60580560, 0.34000283, 1.90663591, 0.34398859, 0.…
$ `2015`  <dbl> 1.50836672, 0.89656529, 0.56142915, 1.12524136, 0.30936455, 0.…
$ `2016`  <dbl> 1.276990945, 0.891592367, 1.973852647, 1.428759547, 0.68350420…
$ `2017`  <dbl> 1.9486474, 2.0812686, 2.1259709, 1.5968841, 2.4505340, 1.14713…
$ `2018`  <dbl> 1.9114009, 1.9983819, 2.0531650, 2.2682257, 2.1494949, 0.81360…
$ `2019`  <dbl> 1.6107679, 1.5308955, 1.4368196, 1.9492690, 2.8478760, 0.75813…
$ `2020`  <dbl> 0.84690554, 1.38190955, 0.74079181, 0.71699963, 3.16129528, 0.…
$ `2021`  <dbl> 2.863910, 2.766667, 2.440249, 3.395193, 3.839845, 1.853045, 2.…
```

country_inflation has 44 rows and each row represents a country. It has 30 columns each representing inflation rate in the country in that year.

b. Display a list of the countries included in the dataset.

```
country_inflation|>pull(country)
```

```
 [1] "Australia"                    "Austria"
 [3] "Belgium"                      "Canada"
 [5] "Czech Republic"               "Denmark"
 [7] "Finland"                      "France"
 [9] "Germany"                      "Greece"
[11] "Hungary"                      "Iceland"
[13] "Ireland"                      "Italy"
[15] "Japan"                        "Korea"
[17] "Luxembourg"                   "Mexico"
[19] "Netherlands"                  "New Zealand"
[21] "Norway"                       "Poland"
[23] "Portugal"                     "Slovak Republic"
[25] "Spain"                        "Sweden"
```

```
 [27] "Switzerland"                 "Türkiye"
 [29] "United Kingdom"              "United States"
 [31] "Argentina"                   "Brazil"
 [33] "Chile"                       "China (People's Republic of)"
 [35] "Estonia"                     "India"
 [37] "Indonesia"                   "Israel"
 [39] "Russia"                      "Saudi Arabia"
 [41] "Slovenia"                    "South Africa"
 [43] "Colombia"                    "Costa Rica"
```

> **Tip**
>
> A function that can be useful for part (b) is `pull()`. Check out its documentation for examples of usage.

## Question 2

Which countries had the top three highest inflation rates in 2021? Your output should be a data frame with two columns, `country` and `2021`, with inflation rates in descending order, and three rows for the top three countries. Briefly comment on how the inflation rates for these countries compare to the inflation rate for United States in that year.

```r
country_inflation |>
  select(country, `2021`) |>
  arrange(desc(`2021`)) |>
  slice(1:3)
```

```
# A tibble: 3 × 2
  country   `2021`
  <chr>      <dbl>
1 Argentina  48.4
2 Türkiye    19.6
3 Brazil      8.30
```

The top three countries with the highest inflation rates in 2021 were Argentina, Turkiye, and Brazil. Argentina had an inflation rate of approximately 48.4%, Turkiye around 19.6%, and Brazil around 8.3%. In comparison, the United States had a relatively moderate inflation rate of about 4.7% in 2021, significantly lower than these three countries, particularly Argentina and Turkey.

> **Tip**
>
> Column names that are numbers are not considered "proper" in R, therefore to select them you'll need to surround them with backticks, e.g. `select( ` 1993 ` )`.

## Question 3

In a single pipeline,

- calculate the ratio of the inflation in 2021 and inflation in 1993 for each country and store this information in a new column called `inf_ratio`,

- arrange the data frame in decreasing order of `inf_ratio`, and
- select the variables `country` and `inf_ratio` to display as the result of the pipeline.

Do not save this new variable in `inf_ratio`, only calculate and display it so you can answer the following question based on the output of the pipeline.

Which country's inflation change is the largest over this time period? Did inflation increase of decrease between 1993 and 2021 in this country?

```
country_inflation |>
  mutate(inf_ratio = `2021` / `1993`) |>
  arrange(desc(inf_ratio)) |>
  select(country, inf_ratio)
```

```
# A tibble: 44 × 2
   country       inf_ratio
   <chr>             <dbl>
 1 New Zealand        3.06
 2 Canada             1.82
 3 Australia          1.63
 4 Ireland            1.60
 5 United States      1.59
 6 Norway             1.52
 7 Denmark            1.47
 8 Iceland            1.10
 9 Netherlands        1.04
10 Finland            1.00
# ℹ 34 more rows
```

The country with the largest inflation change over this time period is New Zealand, with an inflation ratio of approximately 3. This indicates that inflation increased between 1993 and 2021, as the ratio is greater than 1.

> **Tip**
>
> For this question you'll once again need to use variables whose names are numbers (years) in your pipeline. Make sure to surround the names of such variables with backticks (`).

## Question 4

Reshape (pivot) `country_inflation` such that each row represents a country/year combination, with columns `country`, `year`, and `annual_inflation`. Then, display the resulting data frame and state how many rows and columns it has.

Requirements:

- Your code must use one of `pivot_longer()` or `pivot_wider()`. There are other ways you can do this reshaping move in R, but this question requires solving this problem by pivoting.

- In your `pivot_*()` function, you must use `names_transform = as.numeric` as an argument to transform the variable type to numeric as you pivot the data so that in the resulting data frame the year variable is numeric.
- The resulting data frame must be saved as something other than `country_inflation` so you (1) can refer to this data frame later in your analysis and (2) do not overwrite `country_inflation`. Use a short but informative name.

> **Important**
>
> The remaining questions in Part 1 require the use of the pivoted data frame from Question 4.

```
country_inflation_long <- country_inflation |>
  pivot_longer(
    cols = -country,
    names_to = "year",
    values_to = "annual_inflation",
    names_transform = as.numeric
  )

country_inflation_long
```

```
# A tibble: 1,276 × 3
   country    year annual_inflation
   <chr>     <dbl>            <dbl>
 1 Australia  1993            1.75
 2 Australia  1994            1.97
 3 Australia  1995            4.63
 4 Australia  1996            2.62
 5 Australia  1997            0.225
 6 Australia  1998            0.860
 7 Australia  1999            1.48
 8 Australia  2000            4.46
 9 Australia  2001            4.41
10 Australia  2002            2.98
# ℹ 1,266 more rows
```

The reshaped data frame `country_inflation_long` has 1,276 rows and 3 columns: `country`, `year`, and `annual_inflation`.

## Question 5

Use a separate, single pipeline to answer each of the following questions.

Requirement: Your code must use the `filter()` function for each part, not `arrange()`.

a. What is the highest inflation rate observed between 1993 and 2021? The output of the pipeline should be a data frame with one row and three columns. In addition to code and output, your response should include a single sentence stating the country and year.

```
country_inflation_long |>
  filter(annual_inflation == max(annual_inflation, na.rm = TRUE))
```

```
# A tibble: 1 × 3
  country  year annual_inflation
  <chr>   <dbl>            <dbl>
1 Brazil   1994            2076.
```

The highest inflation rate observed was in Brazil in 1994.

b. What is the lowest inflation rate observed between 1993 and 2021? The output of the pipeline should be a data frame with one row and three columns. In addition to code and output, your response should include a single sentence stating the country and year.

```
country_inflation_long |>
  filter(annual_inflation == min(annual_inflation, na.rm = TRUE))
```

```
# A tibble: 1 × 3
  country  year annual_inflation
  <chr>   <dbl>            <dbl>
1 Ireland  2009            -4.48
```

The lowest inflation rate observed was in Ireland in 2009

c. Putting (a) and (b) together: What are the highest and the lowest inflation rates observed between 1993 and 2021? The output of the pipeline should be a data frame with two rows and three columns.

```
country_inflation_long |>
  filter(
    annual_inflation == max(annual_inflation, na.rm = TRUE) |
    annual_inflation == min(annual_inflation, na.rm = TRUE)
  )
```

```
# A tibble: 2 × 3
  country  year annual_inflation
  <chr>   <dbl>            <dbl>
1 Ireland  2009            -4.48
2 Brazil   1994          2076.
```

The highest inflation rate was observed in Brazil in 1994 at approximately 2076% while the lowest inflation rate was observed in Ireland in 2009 at approximately -4.5%.

## Question 6

a. Create a vector called `countries_of_interest` which contains the names of up to five countries you want to visualize the inflation rates for over the years. For example, if these countries are Türkiye and United States, you can express this as follows:

```
countries_of_interest <- c("Türkiye", "United States")
```

If they are Türkiye, United States, and China, you can express this as follows:

```
countries_of_interest <- c(
  "Türkiye", "United States", "China (People's Republic of)"
)
```

So on and so forth... Then, in 1-2 sentences, state why you chose these countries.

> **Note**
>
> Your `countries_of_interest` should consist of no more than five countries. Make sure that the spelling of your countries matches how they appear in the dataset.

```
countries_of_interest <- c(
  "United States",
  "China (People's Republic of)",
  "Japan",
  "Germany",
  "India"
)
countries_of_interest
```

```
[1] "United States"                "China (People's Republic of)"
[3] "Japan"                        "Germany"
[5] "India"
```

I chose these five countries as they represent some of the world's largest economies across continents.

b. In a single pipeline, filter your reshaped dataset to include only the `countries_of_interest` from part (a), and save the resulting data frame with a new name so you (1) can refer to this data frame later in your analysis and (2) do not overwrite the data frame you're starting with. Use a short but informative name. Then, in a new pipeline, find the `distinct()` countries in the data frame you created.

> **Tip**
>
> The number of distinct countries in the filtered data frame you created in part (b) should equal the number of countries you chose in part (a). If it doesn't, you might have misspelled a country name or made a mistake in how to filter for these countries. Go back and check your code.

```
country_inflation_filtered <- country_inflation_long |>
  filter(country %in% countries_of_interest)
country_inflation_filtered
```

```
# A tibble: 145 × 3
   country  year annual_inflation
   <chr>   <dbl>            <dbl>
```

```
 1 Germany  1993            4.47
 2 Germany  1994            2.69
 3 Germany  1995            1.71
 4 Germany  1996            1.45
 5 Germany  1997            1.94
 6 Germany  1998            0.911
 7 Germany  1999            0.585
 8 Germany  2000            1.44
 9 Germany  2001            1.98
10 Germany  2002            1.42
# ℹ 135 more rows
```

```
country_inflation_filtered |>
  distinct(country)
```

```
# A tibble: 5 × 1
  country
  <chr>
1 Germany
2 Japan
3 United States
4 China (People's Republic of)
5 India
```

## Question 7

Using your data frame from the previous question, create a plot of annual inflation vs. year for these countries. Then, in a few sentences, describe the patterns you observe in the plot, particularly focusing on anything you find surprising or not surprising, based on your knowledge (or lack thereof) of these countries economies.

Requirements for the plot:

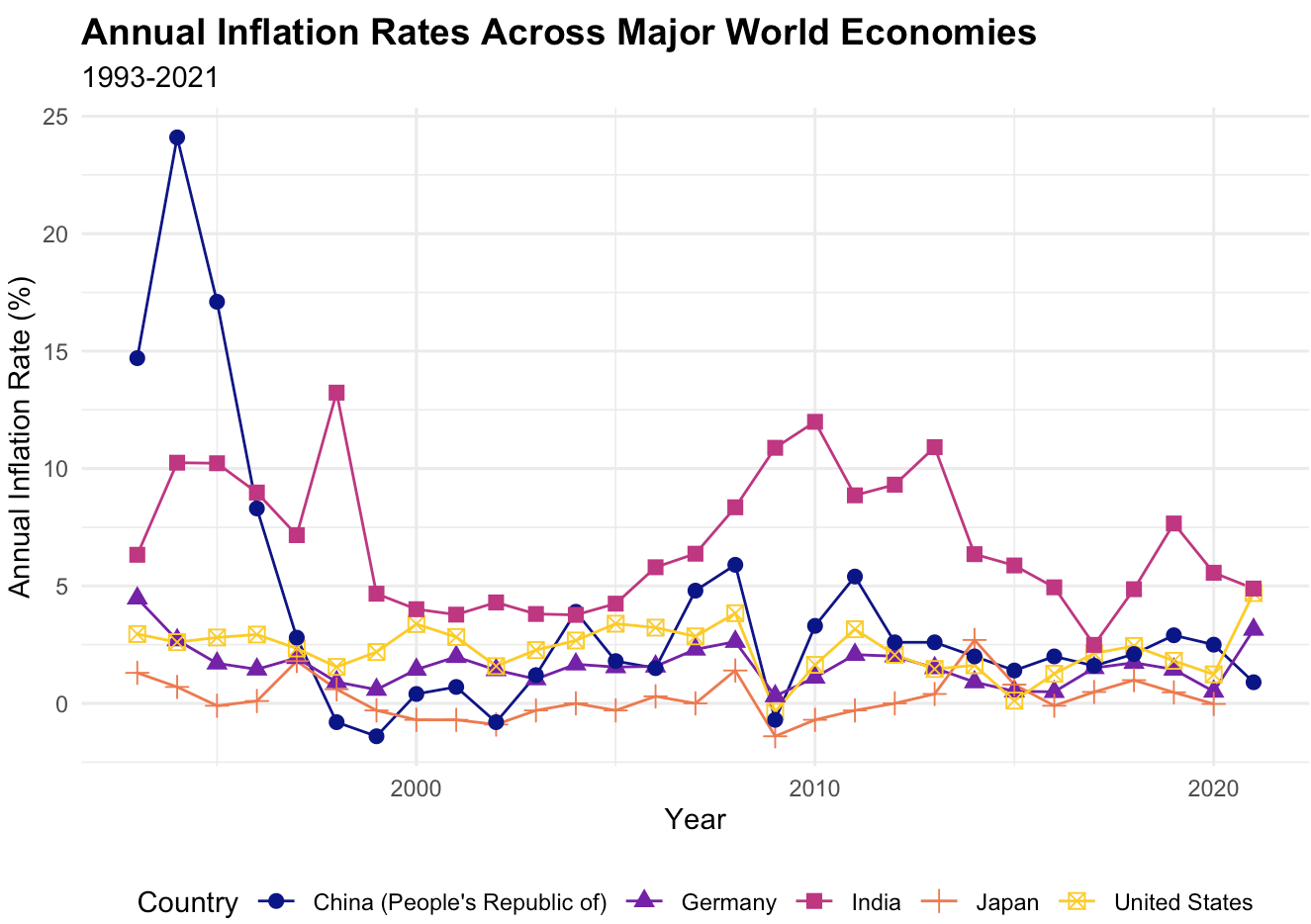
- Data should be represented with points as well as lines connecting the points for each country.
- Each country should be represented by a different color line and different color and shape points.
- Axes and legend should be properly labeled.
- The plot should have an appropriate title (and optionally a subtitle).
- Plot should be customized in at least one way – you could use a different than default color scale, or different than default theme, or some other customization.

```
ggplot(
  country_inflation_filtered,
  aes(x = year, y = annual_inflation, color = country, shape = country)
) +
  geom_line() +
  geom_point(size = 2.5) +
  scale_color_viridis_d(option = "plasma", end = 0.9) +
  labs(
    title = "Annual Inflation Rates Across Major World Economies",
```

```
    subtitle = "1993–2021",
    x = "Year",
    y = "Annual Inflation Rate (%)",
    color = "Country",
    shape = "Country"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    legend.position = "bottom"
  )
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_line()`).

Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```




If you haven't yet done so, now is a good time to render, commit, and push. Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

## Part 2

### Inflation in the US

The OECD defines inflation as follows:

> Inflation is a rise in the general level of prices of goods and services that households acquire for the purpose of consumption in an economy over a period of time.
>
> The main measure of inflation is the annual inflation rate which is the movement of the Consumer Price Index (CPI) from one month/period to the same month/period of the previous year expressed as percentage over time.
>
> Source: [OECD CPI FAQ]

CPI is broken down into 12 divisions such as food, housing, health, etc. Your goal in this part is to create another time series plot of annual inflation, this time for US only.

The data you will need to create this visualization is spread across two files:

- `us-inflation.csv`: Annual inflation rate for the US for 12 CPI divisions. Each division is identified by an ID number.
- `cpi-divisions.csv`: A "lookup table" of CPI division ID numbers and their descriptions.

Let's load both of these files.

```
us_inflation <- read_csv("data/us-inflation.csv")
cpi_divisions <- read_csv("data/cpi-divisions.csv")
```

## Question 8

a. How many columns and how many rows does the `us_inflation` dataset have? What are the variables in it? Add a brief (1-2 sentences) narrative summarizing this information.

```
glimpse(us_inflation)
```

```
Rows: 132
Columns: 4
$ country          <chr> "United States", "United States", "United States", "U…
$ cpi_division_id  <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2,…
$ year             <dbl> 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019,…
$ annual_inflation <dbl> 4.8039240, 2.4539620, 0.9083796, 2.6398010, 1.1677110…
```

The `us_inflation` dataset has 4 columns and 132 rows . The variables are: `country` (always "United States"), `cpi_division_id` (a numeric identifier from 1-12), `year` (the year of observation), and `annual_inflation` (the annual inflation rate for that division and year).

b. How many columns and how many rows does the `cpi_divisions` dataset have? What are the variables in it? Add a brief (1-2 sentences) narrative summarizing this information.

```
glimpse(cpi_divisions
```

```
)
```

```
Rows: 12
Columns: 2
$ id          <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
$ description <chr> "Food and non-Alcoholic beverages", "Alcoholic beverages, …
```

The `cpi_divisions` dataset has 2 columns and 12 rows. The variables are: `id` (a numeric identifier from 1-12 matching the CPI division codes) and `description` (a text description of what each division represents, such as "Food and non-Alcoholic beverages".

c. Create a new dataset by joining the `us_inflation` dataset with the `cpi_division_id` dataset.

- Determine which type of join is the most appropriate one and use that.
- Note that the two datasets don't have a common variable. Review the help for the join functions to determine how to use the `by` argument when the names of the variables that the datasets should be joined by are different.
- Use a short but informative name for the joined dataset, and do not overwrite either of the datasets that go into creating it.

Then, find the number of rows and columns of the resulting dataset and report the names of its columns. Add a brief (1-2 sentences) narrative summarizing this information.

```
us_inflation_joined <- us_inflation |>
  left_join(cpi_divisions, by = c("cpi_division_id" = "id"))

glimpse(us_inflation_joined)
```

```
Rows: 132
Columns: 5
$ country          <chr> "United States", "United States", "United States", "U…
$ cpi_division_id  <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2,…
$ year             <dbl> 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019,…
$ annual_inflation <dbl> 4.8039240, 2.4539620, 0.9083796, 2.6398010, 1.1677110…
$ description      <chr> "Food and non-Alcoholic beverages", "Food and non-Alc…
```

The joined dataset `us_inflation_joined` has 5 columns and 132 rows. The columns are: `country`, `cpi_division_id`, `year`, `annual_inflation`, and `description`. A left join was most appropriate because we want to keep all the inflation data and add the descriptive labels from `cpi_divisions`.

## Question 9

a. Create a vector called `divisions_of_interest` which contains the descriptions or IDs of CPI divisions you want to visualize. Your `divisions_of_interest` should consist of no more than five divisions. If you're using descriptions, make sure that the spelling of your divisions matches how they appear in the dataset. Then, in 1-2 sentences, state why you chose these divisions.

> **Tip**

> Refer back to the guidance provided in Question 6 if you're not sure how to create this vector.

```
divisions_of_interest <- c(
  "Food and non-Alcoholic beverages",
  "Housing, water, electricity, gas and other fuels",
  "Transport",
  "Health",
  "Education"
)
```

I chose these five divisions because they represent essential spending categories that most households face.

b. In a single pipeline, filter your reshaped dataset to include only the `divisions_of_interest` from part (a), and save the resulting data frame with a new name so you (1) can refer to this data frame later in your analysis and (2) do not overwrite the data frame you're starting with. Use a short but informative name. Then, in a new pipeline, find the `distinct()` divisions in the data frame you created.

```
us_inflation_filtered <- us_inflation_joined |>
  filter(description %in% divisions_of_interest)
us_inflation_filtered
```

```
# A tibble: 55 × 5
   country       cpi_division_id  year annual_inflation description
   <chr>                   <dbl> <dbl>            <dbl> <chr>
 1 United States               1  2011            4.80  Food and non-Alcoholic …
 2 United States               1  2012            2.45  Food and non-Alcoholic …
 3 United States               1  2013            0.908 Food and non-Alcoholic …
 4 United States               1  2014            2.64  Food and non-Alcoholic …
 5 United States               1  2015            1.17  Food and non-Alcoholic …
 6 United States               1  2016           -1.33  Food and non-Alcoholic …
 7 United States               1  2017           -0.202 Food and non-Alcoholic …
 8 United States               1  2018            0.456 Food and non-Alcoholic …
 9 United States               1  2019            0.885 Food and non-Alcoholic …
10 United States               1  2020            3.51  Food and non-Alcoholic …
# ℹ 45 more rows
```

```
us_inflation_filtered |>
  distinct(description)
```

```
# A tibble: 5 × 1
  description
  <chr>
1 Food and non-Alcoholic beverages
2 Housing, water, electricity, gas and other fuels
3 Health
```

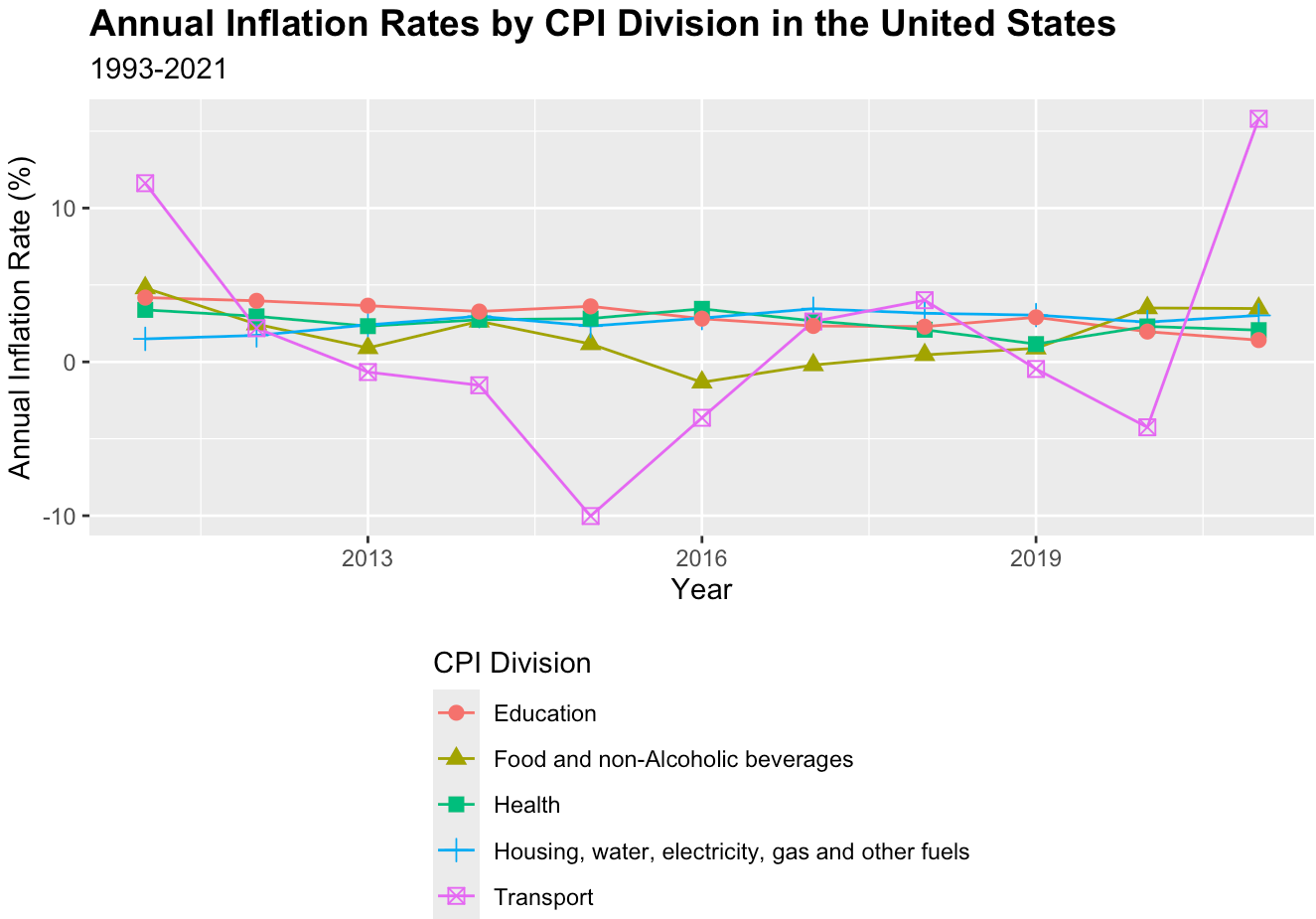
```
4 Transport
5 Education
```

## Question 10

Using your data frame from the previous question, create a plot of annual inflation vs. year for these divisions. Then, in a few sentences, describe the patterns you observe in the plot, particularly focusing on anything you find surprising or not surprising, based on your knowledge (or lack thereof) of inflation rates in the US over the last decade.

- Data should be represented with points as well as lines connecting the points for each division.
- Each division should be represented by a different color line and different color and shape points.
- Axes and legend should be properly labeled.
- The plot should have an appropriate title (and optionally a subtitle).
- Plot should be customized in at least one way – you could use a different than default color scale, or different than default theme, or some other customization.
- If your legend has labels that are too long, you can try moving the legend to the bottom and stack the labels vertically. *Hint:* The `legend.position` and `legend.direction` arguments of the `theme()` functions will be useful.

```r
ggplot(
  us_inflation_filtered,
  aes(x = year, y = annual_inflation, color = description, shape = description)
) +
  geom_line() +
  geom_point(size = 2.5) +
  labs(
    title = "Annual Inflation Rates by CPI Division in the United States",
    subtitle = "1993-2021",
    x = "Year",
    y = "Annual Inflation Rate (%)",
    color = "CPI Division",
    shape = "CPI Division"
  ) +
  theme_gray() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    legend.position = "bottom",
    legend.direction = "vertical"
  )
```




If you haven't yet done so since Part 1, now is a good time to render, commit, and push. Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

# Wrap-up

## Submission

Once you are finished with the lab, you will submit your final PDF document to Canvas.

> **Warning**
>
> Before you wrap up the assignment, make sure all of your documents are updated on your GitHub repo. We will be checking these to make sure you have been practicing how to commit and push changes.
>
> You must turn in a PDF file to the Canvas page by the submission deadline to be considered "on time".

To submit your assignment:

- Go to http://www.Canvas.com and click *Log in* in the top right corner.
- Click *School Credentials* → *Duke NetID* and log in using your NetID credentials.
- Click on your *REN R 213* course.
- Click on the assignment, and you'll be prompted to submit it.

- Mark all the pages associated with question. All the pages of your lab should be associated with at least one question (i.e., should be “checked”).

**Checklist**

Make sure you have:

- attempted all questions
- rendered your Quarto document
- committed and pushed everything to your GitHub repository such that the Git pane in RStudio is empty
- uploaded your PDF to Canvas
- selected pages associated with each question on Canvas

## Grading

The lab is graded out of a total of 36 points.

You can earn up to 4 points on each question:

- 4: Response shows excellent understanding and addresses all or almost all of the rubric items.
- 3: Response shows good understanding and addresses most of the rubric items.
- 2: Response shows understanding and addresses a majority of the rubric items.
- 1: Response shows effort and misses many of the rubric items.
- 0: Response does not show sufficient effort or understanding and/or is largely incomplete.

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### Footnotes

1. Remember, haikus not novellas when writing code! ↩